

Ann prac4

```
import numpy as np
```

```
# Input patterns (binary data)
```

```
patterns = np.array([
```

```
    [1, 0, 1, 0],
```

```
    [1, 1, 0, 0],
```

```
    [0, 0, 1, 1],
```

```
    [1, 0, 1, 1]
```

```
])
```

```
# Vigilance parameter
```

```
vigilance = 0.5
```

```
# Cluster list
```

```
clusters = []
```

```
# ART-like clustering
```

```
for p in patterns:
```

```
    matched = False
```

```
    for c in clusters:
```

```
        # Similarity measure
```

```
        similarity = np.sum(p & c) / np.sum(p)
```

```
        if similarity >= vigilance:
```

```
            matched = True
```

```
            break
```

```
    if not matched:
```

```
        clusters.append(p)
```

# Output

```
print("Number of clusters formed:", len(clusters))
```

```
print("Clusters:")
```

```
for i, c in enumerate(clusters):
```

```
    print(f"Cluster {i+1}: {c}")
```